

Azamat Turganbayev

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Education

University of Minnesota (UMN), Twin Cities

MS in Robotics (Full Scholarship, GPA: 3.89/4.00)

Sep 2025 - Dec 2026

Twin Cities, MN

- Coursework: Machine Learning Fundamentals, Robotics, Robot Vision

Nazarbayev University (NU, Leading University in Kazakhstan)

Aug 2018 - Jun 2023

BS in Robotics and Mechatronics

Astana, Kazakhstan

- Graduation Project: Shoulder Rehabilitation Robotic Exoskeleton
- Coursework: Electromechanical Systems, Mechanical Design with CAD, Image Processing

Skills

Programming Languages & Libraries: Python, C/C++, MATLAB, OpenCV, NumPy

Robotics Frameworks & Tools: ROS/ROS2, Gazebo, Linux

Modeling & Simulation: SolidWorks, MATLAB/Simulink, QUARC

Control Systems: Kinematics, P/PI Controller Tuning, Real-Time Systems

Work Experience

Senior Research Assistant

Nov 2023 - Apr 2025

Center of Excellence in Medical Robotics & Research, NU

Astana, Kazakhstan

- Contributed to the integration of control algorithms with MATLAB/Simulink and QUARC

Robotics Laboratory Assistant

Nov 2023 - Jul 2025

School of Engineering and Digital Sciences, NU

Astana, Kazakhstan

- Developed and co-led experimental setups to demonstrate robotic control concepts

Undergraduate Research Assistant

Apr 2022 - Nov 2023

School of Engineering and Digital Sciences, NU

Astana, Kazakhstan

- Designed and built a 3D-printed prototype of a robotic shoulder exoskeleton

Research & Project Experience

ArUco Marker-Based Navigation for TurtleBot3 Burger (ROS 2, OpenCV, Python)

Oct 2025 - Dec 2025

College of Science & Engineering, UMN

Twin Cities, MN

- Integrated and calibrated Raspberry Pi camera for ArUco marker-based tracking
- Fused vision and LiDAR data to maintain a precise 10 cm target following distance

Obstacle Avoidance for TurtleBot3 Burger (ROS 2, Python)

Oct 2025 - Dec 2025

College of Science & Engineering, UMN

Twin Cities, MN

- Built a ROS2 pipeline to detect hazards and execute reactive maneuvers within 20 cm
- Filtered sensor noise to bridge the gap between Gazebo simulation and physical hardware

Autonomous Maze Navigation with TurtleBot3 Burger (ROS, OpenCV, Python)

Mar 2023 - Jul 2025

School of Engineering and Digital Sciences, NU

Astana, Kazakhstan

- Implemented vision-based localization and control for maze navigation
- Performed LiDAR-based SLAM for maze mapping

Real-Scale Robotic Shoulder Exoskeleton (SolidWorks, MATLAB/Simulink, QUARC)

Nov 2023 - Apr 2025

School of Engineering and Digital Sciences, NU

Astana, Kazakhstan

- Built a full-scale 5-DOF shoulder rehabilitation exoskeleton
- Conducted P/PI controller tuning for linear and rotational drives

3D-Printed Miniature Shoulder Exoskeleton Prototype (SolidWorks, MATLAB)

Apr 2022 - Nov 2023

School of Engineering and Digital Sciences, NU

Astana, Kazakhstan

- Designed a miniature robotic shoulder exoskeleton for rehabilitation applications
- Performed MATLAB/CAD kinematic verification of the exoskeleton

Publications & Conferences

- A. Niyetkaliyev, A. Turganbayev, M. Karasheva, R. Zhylkaidarov and Y. Turlybek, "Exploring the Potential of Four-Bar Linkages in Robotic Exoskeletons: A Comprehensive Review," in *Journal of Mechanical Design*, vol. 147, no. 10, April 2025, doi: [10.1115/1.4068107](https://doi.org/10.1115/1.4068107).
- M. Karasheva, A. Turganbayev, A. Aimysheva and A. Niyetkaliyev, "Design of a 3D Printed Miniature Model for Human-Robot Mechanism Coupling for Shoulder Rehabilitation," *2023 8th International Conference on Robotics and Automation Engineering (ICRAE)*, Singapore, Singapore, 2023, pp. 58-65, doi: [10.1109/ICRAE59816.2023.10458641](https://doi.org/10.1109/ICRAE59816.2023.10458641).
- S. Omirbayev, I. Issa, Z. Kuangaliyev, A. Turganbayev and A. Niyetkaliyev, "The Use of Four-Bar Mechanisms in Robotic Exoskeletons," *2022 International Conference on Advanced Mechatronic Systems (ICAMechS)*, Toyama, Japan, 2022, pp. 149-156, doi: [10.1109/ICAMechS57222.2022.10003280](https://doi.org/10.1109/ICAMechS57222.2022.10003280).